

1. (Use an 8 page booklet)

[Marks 1, 2, 2, 2, 2, 3]

- (a) A coin is tossed at the same time that a die is thrown. What is the probability of getting a 3 on the die and a head on the coin?
- (b) Factor  $3x^2 - x - 4$
- (c) Find the exact value of  $\operatorname{cosec} 60^\circ + \tan 30^\circ$
- (d) Find integers  $a$  and  $b$  such that  $(3 - \sqrt{5})^2 = a + b\sqrt{5}$
- (e) Solve  $|2x - 1| = 5$
- (f) Solve the simultaneous equations:  $3x - y = 5$   
 $5x + 3y = -8$

2. (Use the same booklet as for question 1)

[Marks 2, 5, 5]

- (a) A die is rolled  $n$  times. Find an expression to describe the probability of rolling  $n$  sixes.
- (b) In a certain factory a machine manufactures bicycle components, of which 3% are faulty. I buy 3 such components. By drawing a tree diagram, or otherwise, find the probability that:
- (i) exactly one component is faulty.
- (ii) at least one is faulty.
- (c) A ball is dropped from a height of 3 m on to the floor. After each bounce the maximum height reached by the ball is 60% of the previous maximum height. Thus, after the first bounce it reaches a height of 180 cm, before falling again.
- (i) What is the height reached after the second bounce?
- (ii) What kind of sequence is formed by the successive heights?
- (iii) What is the total distance travelled by the ball from the time it was dropped until it comes to rest?

3. (Use a new 4 page booklet)

[Marks 2, 2, 3, 2, 1, 2]

If A, B and C are the points (5, 3), (-2, 5) and (4, -3) respectively, find the:

- (a) exact distance from A to C.
- (b) gradient of the join of A and C.
- (c) equation of the line passing through A and C, in general form.
- (d) perpendicular distance from B to the line passing through A and C.
- (e) area of the triangle ABC.
- (f) co-ordinates of the point D such that ABCD is a parallelogram.

4. (Use a new 4 page booklet)

[Marks 4, 3, 5]

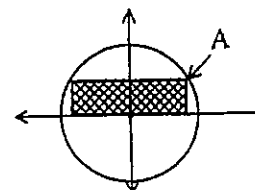
- (a) For the equation  $x^2 + (m-3)x + m = 0$  find the values of  $m$  for which the equation has
- (i) two equal roots
- (ii) two distinct real roots
- (b) Find all real roots of the equation  $4^x - 6 \times 2^x + 8 = 0$
- (c) For the parabola  $2y^2 - 4x - 8 = 0$ , find
- (i) the co-ordinates of the vertex
- (ii) the co-ordinates of the focus
- (iii) the equation of the directrix

5. (Use a new 8 page booklet)

[Marks 2, 4, 6]

- (a) Differentiate  $\log_e \left( \frac{2x-3}{x+1} \right)$  with respect to  $x$  leaving your answer without simplifying it.
- (b) Show that the graph of  $y = e^x \cos x$  has a stationary point at  $x = \frac{\pi}{4}$  and determine its nature.

(c)

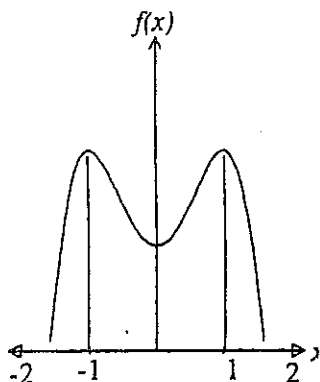
The diagram shows a rectangle inscribed in a semicircle whose equation is  $y = \sqrt{4 - x^2}$ .

- (i) Let the  $x$  coordinate of point A be  $x$ . Show the area of the rectangle is given by  $2x\sqrt{4 - x^2}$ .
- (ii) Find the maximum area of the rectangle.

6. (Use the same booklet as for question 5)

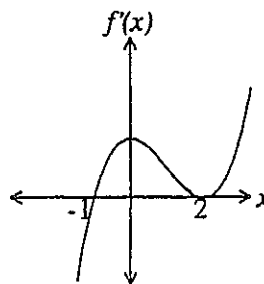
[Marks 5, 4, 3]

- (a) The graph shows part of the function  $y = 2\cos(1-x^2)$ . Using 5 function values, find an approximation using the trapezoidal rule for the area between this graph, the  $x$ -axis and the lines  $x = 1$  and  $x = -1$ .



- (b) The area enclosed by the graph of  $y = \frac{1}{\sqrt{1-x}}$ , the coordinate axes and the line  $x = \frac{1}{2}$  is rotated around the  $x$  axis. Calculate the volume of the solid so formed.

- (c) The diagram below shows the graph of the derived function  $f'(x)$ , for the function  $y = f(x)$ .



- (i) Explain the behaviour of the graph of  $y = f(x)$  at  $x = -1$  and at  $x = 2$ .  
 (ii) Explain what you know about the graph of  $y = f(x)$  when  $x$  is between these values.

7. (Use a new 8 page booklet)

[Marks 2, 2, 2, 4, 2]

- (a) Find the derivatives, without simplifying, of each of:

(i)  $e^{\cos 3x}$  (ii)  $\sqrt{xe^{2x}}$  (iii)  $\tan^3(5x+4)$

- (b) (i) Sketch, on the same set of axes, graphs of  $y = 3\sin 2x$  and  $y = \frac{x}{2}$   
 (ii) State the number of solutions of the equation  $6\sin 2x - x = 0$

8. (Use the same booklet as for question 7)

[Marks 3, 3, 3, 3]

- (a) Show that the graph of  $y = \log_e x$  is concave down and increasing for all values of  $x > 0$   
 (b) Show that if  $y = A \cos 4x + B \sin 4x$  then  $\frac{d^2 y}{dx^2} + 16y = 0$ .  
 (c) The area under the curve  $y = \frac{1}{x}$  between  $x = 1$  and  $x = b$  is equal to 1 square unit. What is the value of  $b$ ?  
 (d) If the gradient of the tangent at the point  $(x, y)$  on a particular curve is  $2\sin 3x$  and the curve passes through the point  $(\frac{\pi}{3}, \frac{8}{3})$ , find the equation of the curve.

9. (Use a new 8 page booklet)

[Marks 1, 2, 2, 3, 4]

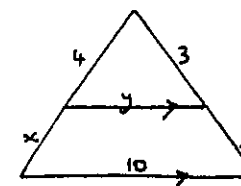
- (a) Given  $F(x) = \frac{x+1}{x-1}$ ,  $G(x) = x^2$ ,  $H(x) = \sqrt{x}$

- (i) State the domain of  $F(x)$ .  
 (ii) Evaluate, where possible,  $HG(-3)$  and  $GH(-3)$ .  
 [If either is not possible, explain why.]

- (iii) Show that  $F\left(\frac{1}{x}\right) = -F(x)$ .

- (iv) Show that  $F^{-1}(x) = F(x)$ .

- (b) From the diagram evaluate  $x$  and  $y$ .



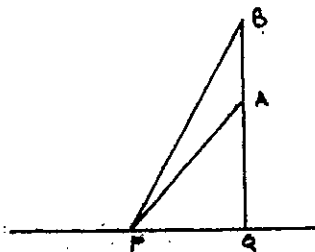
10. (Use the same booklet as for question 9)

[Marks 4, 3, 5]

- (a) A point  $P(x, y)$  moves so that its distance from  $(-1, 1)$  is equal to its distance from the straight line  $3x - 4y + 4 = 0$ .

Show that the cartesian equation of the locus is  $16x^2 + 26x + 24xy + 9y^2 - 18y + 34 = 0$ .

- (b) AB is a 12 m flag pole on the top of a building. Find the length of BP given that  $\angle QPB = 56^\circ 37'$  and  $\angle QPA = 43^\circ 19'$ .



- (c) ABCD is a parallelogram. J and K are points of trisection of the diagonal AC (i.e.  $AJ = JK = KC$ ).

(i) Draw a clear diagram to represent this situation.

(ii) Prove that  $\triangle ADJ \cong \triangle CBK$ .

(iii) Hence explain why  $DJ \parallel BK$ .

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1} \quad (n \neq -1; x \neq 0 \text{ if } n < 0)$$

$$\int \frac{1}{x} dx = \log_e x \quad (x > 0)$$

$$\int \cos ax dx = \frac{1}{a} \sin ax \quad (a \neq 0)$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax \quad (a \neq 0)$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a} \quad (a \neq 0)$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a} \quad (a > 0, -a < x < a)$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \log_e \left\{ x + \sqrt{x^2 - a^2} \right\} \quad (|x| > |a|)$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \log_e \left\{ x + \sqrt{x^2 + a^2} \right\}$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax} \quad (a \neq 0)$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax \quad (a \neq 0)$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax \quad (a \neq 0)$$

Question 1

(a)  $P(3) = \frac{1}{6}$   $P(H) = \frac{1}{2}$

$P(3 \text{ AND } H) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$

(b) Factorize  $3x^2 - x - 4 = (3x - 4)(x + 1)$

$3x - 4$   
 $x + 1$

(c)  $\csc 60^\circ + \tan 30^\circ = \frac{1}{\sin 60^\circ} + \tan 30^\circ$

$= \frac{1}{\frac{\sqrt{3}}{2}} + \frac{1}{\sqrt{3}}$

$= \frac{2}{\sqrt{3}} + \frac{1}{\sqrt{3}}$

$= \frac{3}{\sqrt{3}}$

$= \sqrt{3}$

$= \boxed{\sqrt{3}}$

(d)  $(3 - \sqrt{5})^2 = (3)^2 - 2(3)(\sqrt{5}) + (\sqrt{5})^2$

$= 9 - 6\sqrt{5} + 5$

$= 14 - 6\sqrt{5}$

$\therefore a = 14$   $b = -6$

(e)  $|2x - 1| = 5$

$2x - 1 = 5$  OR  $-(2x - 1) = 5$

$2x = 6$   $2x - 1 = -5$

$x = 3$   $2x = -4$

$x = -2$

Check Solutions

$\therefore x = 3$  AND  $x = -2$  ARE Solutions.

(f) Solving

$3x - y = 5 \dots (1)$

$5x + 3y = -8 \dots (2)$

Elimination

$(1) \times 3$   $9x - 3y = 15 \dots (3)$

$5x + 3y = -8 \dots (2) +$

$14x = 7$

$\therefore x = \frac{1}{2}$

Sub  $x = \frac{1}{2}$  into eqn (1) to find  $y$

ie  $3(\frac{1}{2}) - y = 5$

$\frac{3}{2} - y = 5$

$y = -5 + \frac{3}{2}$

$y = \frac{-10 + 3}{2}$

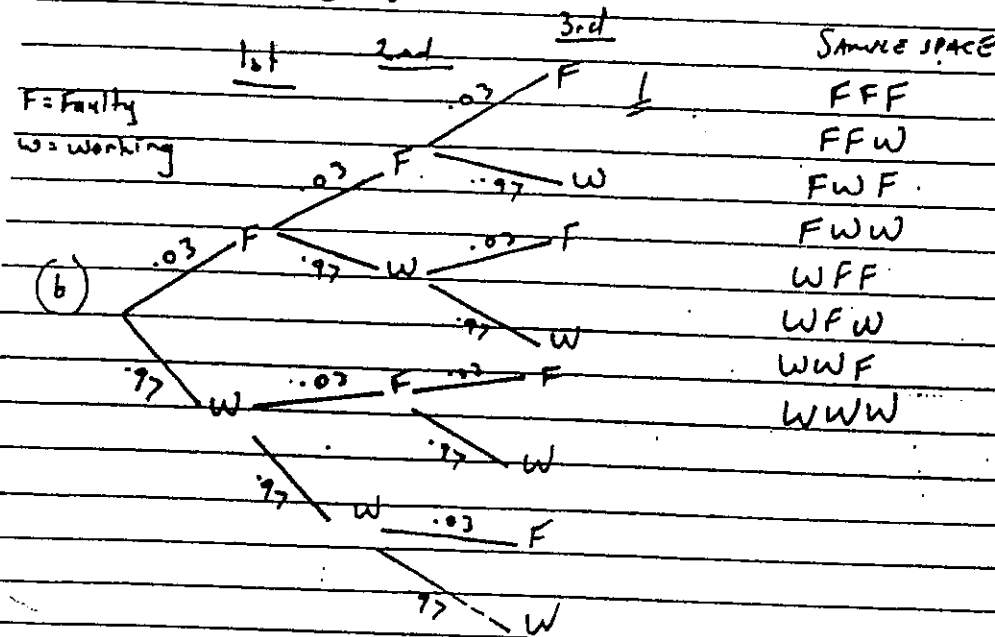
$\therefore y = -\frac{7}{2}$

Hence Solution to Simult equations

is  $(\frac{1}{2}, -\frac{7}{2})$

Question 2

(a)  $(\frac{1}{6})^n$



(i)  $P(\text{EXACTLY ONE Faulty}) = P(FWW) + P(WFW) + P(WWF)$

$= (0.03 \times 0.97 \times 0.97) + (0.97 \times 0.03 \times 0.97) + (0.97 \times 0.97 \times 0.03)$

$= (0.028) + (0.028) + (0.028)$

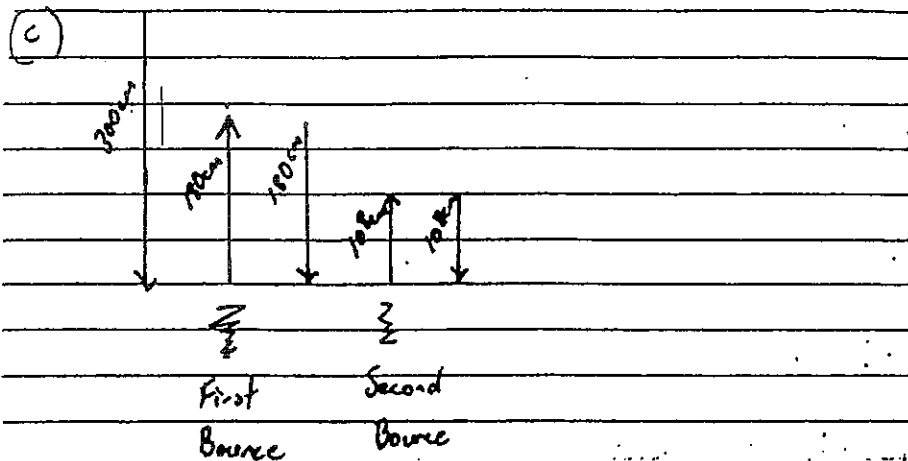
$= 0.084$

(ii)  $P(\text{at least one Faulty}) = 1 - P(WWW)$

$= 1 - (0.97 \times 0.97 \times 0.97)$

$= 1 - 0.913$

$= 0.087$



(i)  $1.8m$  or  $108cm$   $\frac{1}{2}$

(iii) accept Infinite Sequence or Geometric or limiting height

(iii) Total Distance =  $3 + 2(1.8 + 1.08 + 0.648)$   $\frac{1}{2}$

$$= 3 + 2 \left( \frac{a}{1-r} \right)$$

$$= 3 + 2 \left( \frac{1.8}{1-0.6} \right) \frac{1}{2}$$

$$= 3 + 9$$

$\approx$

So Ball travels =  $12m$   $\frac{1}{2}$  or  $1200cm$

Q3.  $A(5,3)$ ,  $B(-2,5)$ ,  $C(4,-3)$

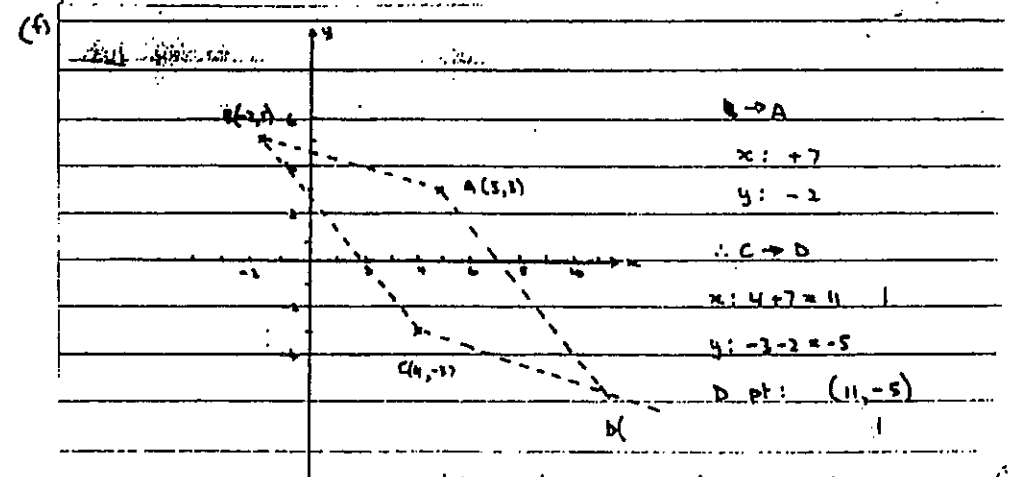
(a)  $d_{AC} = \sqrt{(5-4)^2 + (3-(-3))^2}$   
 $= \sqrt{1+36} = \sqrt{37}$

(b)  $m_{AC} = \frac{6}{1} = 6$

(c)  $y-3 = 6(x-5)$   $|$   $y-3 = 6x-30$   
 $6x - y - 27 = 0$

(d)  $d = \frac{|6(-2) - 5 - 27|}{\sqrt{6^2 + 1}} = \frac{44}{\sqrt{37}}$

(e)  $A = \frac{1}{2} \cdot \frac{44}{\sqrt{37}} \cdot \sqrt{37} = 22 \text{ u}^2$



Note:  $(-3, -1)$  gives a point in the parallelogram ABCD. You were asked for ABCD.

4. (a)  $\Delta = (n-3)^2 - 4m$   
 $= n^2 - 10n + 9$

(i) Two equal roots if  $n^2 - 10n + 9 = 0$   
 $(n-9)(n-1) = 0$   
 $n = 1, 9$

(ii) Two distinct roots if  $n^2 - 10n + 9 > 0$   
 $n < 1$  or  $n > 9$

(b)  $4^x - 6 \times 2^x + 8 = 0$   
 $(2^x - 2)(2^x - 4) = 0$   
 $\therefore 2^x = 2, 4$   
 $x = 1, 2$

(c)  $2y^2 - 5x - 8 = 0$   
 $y^2 = 2(x+2)$   
 (i) Vertex =  $(-2, 0)$   
 (ii) Focus =  $(-1, 0)$   
 (iii) Directrix  $x = -3$

131

QUESTION 5

(a) Outcomes assessed: H3, P7

Marking Criteria

| Criteria                                       | Marks |
|------------------------------------------------|-------|
| • 1 mark for correct differentiation procedure | 2     |
| • 1 mark for correct answer                    |       |

Answer

$$y = \ln\left(\frac{2x-3}{x+1}\right)$$

$$= \ln(2x-3) - \ln(x+1)$$

$$\therefore \frac{dy}{dx} = \frac{2}{2x-3} - \frac{1}{x+1} \quad \text{OR} \quad \frac{x+1}{2x-3} \times \frac{2(x-1) - (2x+3)}{(x+1)^2}$$

(b) Outcomes assessed: H5, P6, P7

| Criteria                                                          | Marks |
|-------------------------------------------------------------------|-------|
| • 1 mark for derivative                                           | 4     |
| • 1 mark for showing derivative 0 at $x = \frac{\pi}{4}$          |       |
| • 1 mark for either sign diag for derivative or second derivative |       |
| • 1 mark for correct conclusion from the above                    |       |

Answer

$$y = e^x \cos x$$

$$y' = e^x \cos x - e^x \sin x$$

$$\text{at } x = \frac{\pi}{4}, y' = e^{\frac{\pi}{4}} \left( \frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) = 0$$

$$\therefore \text{stationary pt at } x = \frac{\pi}{4}$$

$$y'' = e^x (\cos x - \sin x) + e^x (-\sin x - \cos x) = -2e^x \sin x$$

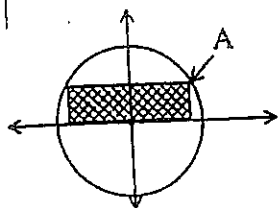
$$\text{at } x = \frac{\pi}{4}, y'' = -2e^{\frac{\pi}{4}} \times \frac{1}{\sqrt{2}} < 0$$

$$\therefore \text{maximum turning point}$$

(c) Outcomes assessed: H1, H2, H5

| Criteria                                         | Marks |
|--------------------------------------------------|-------|
| • 1 mark for identifying dimensions of rectangle | 6     |
| • 1 mark for area                                |       |
| • 1 mark for derivative of area                  |       |
| • 1 mark for solution of $a' = 0$                |       |
| • 1 mark for justifying maximum                  |       |
| • 1 mark for max area = 4                        |       |

Answer:



A's coordinates are  $(x, \sqrt{4-x^2})$

Thus dimensions of rectangle are  $2x \times \sqrt{4-x^2}$

ie, area =  $2x\sqrt{4-x^2}$

$$a = 2x\sqrt{4-x^2}$$

$$\frac{da}{dx} = 2 \times \sqrt{4-x^2} + 2x \times \frac{1}{\sqrt{4-x^2}} \times -2x$$

$$= \frac{8-4x^2}{\sqrt{4-x^2}}$$

$$= 0 \text{ if } x^2 = 2$$

$$x = \sqrt{2} \text{ (must be positive)}$$

$$\frac{da}{dx} > 0 \text{ if } x < \sqrt{2} \text{ and } \frac{da}{dx} < 0 \text{ if } x > \sqrt{2}$$

ie, max turning point

$$\text{and } a_{\max} = 2\sqrt{2} \times \sqrt{2} = 4$$

#### QUESTION 6

(a) Outcomes assessed: P3, P4, H5,

| Criteria                                                                                                                                                                                                     | Marks |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>2 marks for correct function values</li> <li>2 marks for substituting correctly into a correct formula</li> <li>1 mark for evaluation from correct formula</li> </ul> | 5     |

Answer:

|   |    |          |          |          |   |
|---|----|----------|----------|----------|---|
| x | -1 | -0.5     | 0        | 0.5      | 1 |
| y | 2  | 1.463... | 1.081... | 1.463... | 2 |

$$\text{area} = \frac{0.5}{2} (2 + 2 + 2(1.463 + 1.081 + 1.463))$$

$$= 3.0036...$$

$$= 3.00 \text{ units of area (2d.p.)}$$

(b) Outcomes assessed: P3, H8

| Criteria                                                                                                                                                                                                   | Marks |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>1 mark for correct volume formula</li> <li>1 mark for integral of correct x function</li> <li>1 mark for correct integral</li> <li>1 mark for evaluation</li> </ul> | 4     |

Answer

$$\begin{aligned}
 V &= \pi \int_0^{\frac{1}{2}} y^2 dx \\
 &= \pi \int_0^{\frac{1}{2}} \frac{1}{1-x} dx \\
 &= \pi [-\ln(1-x)]_0^{\frac{1}{2}} \\
 &= \pi \ln 2
 \end{aligned}$$

(c) Outcomes assessed: P7, P8, H2, H5, H6, H7

| Criteria                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| <ul style="list-style-type: none"> <li>1 mark for identifying the minimum at <math>x = -1</math></li> <li>1 mark for identifying the stationary inflexion at <math>x = 2</math><br/>OR 1 mark for identifying that there is a stationary point at each of <math>x = -1</math> and <math>x = 2</math></li> <li>1 mark for identifying the inflexion between <math>x = -1</math> and <math>x = 2</math> OR identifying that <math>f(x)</math> is increasing between these values</li> </ul> | 3     |

Answer:

(i)

$$\text{at } x = -1 \quad f'(x) = 0$$

$$f'(-1^-) < 0$$

$$f'(-1^+) > 0$$

$\therefore$  minimum turning point

$$\text{at } x = 2, \quad f'(x) = 0$$

$$f'(2^+) > 0 \text{ and } f'(2^-) > 0$$

$\therefore$  stationary inflexion

(ii)

$$\text{for } -1 < x < 2 \quad f'(x) > 0$$

$\therefore f(x)$  is increasing

also, in this interval the gradient of  $f'(x)$  changes from positive to negative, so there

is a of cur ie, an inflexion at  $x = 2$

QUESTION 7

(a) (i)

Outcomes assessed: P7 H3

Marking Criteria

| Criteria                                       | Marks |
|------------------------------------------------|-------|
| • 1 mark for correct differentiation procedure | 2     |
| • 1 mark for correct answer                    |       |

Answer

$$\frac{d}{dx} e^{\cos 3x} = e^{\cos 3x} \times -\sin 3x \times 3$$

(ii) Outcomes assessed: P7

Marking Criteria

| Criteria                                       | Marks |
|------------------------------------------------|-------|
| • 1 mark for correct differentiation procedure | 2     |
| • 1 mark for correct answer                    |       |

Answer

$$\begin{aligned} \text{(iv)} \quad \frac{d}{dx} \sqrt{x} e^x &= x^{\frac{1}{2}} (e^x)^{\frac{1}{2}} \quad \text{or} \quad \frac{d}{dx} (x e^{2x})^{\frac{1}{2}} \\ &= \frac{1}{2} (x e^{2x})^{-\frac{1}{2}} \times [x e^{2x} + e^{2x}] \\ &= \frac{1}{2} \cdot \frac{1}{\sqrt{x}} \cdot \frac{1}{e} \times e^{2x} (2x+1) \\ &= e^x \left( \frac{1}{2\sqrt{x}} + \frac{1}{2\sqrt{x}} \right) \\ &= e^x \left( \frac{1}{\sqrt{x}} + \frac{1}{2\sqrt{x}} \right) \approx \text{before} \end{aligned}$$

(iii) Outcomes assessed: P7

Marking Criteria

| Criteria                                       | Marks |
|------------------------------------------------|-------|
| • 1 mark for correct differentiation procedure | 2     |
| • 1 mark for correct answer                    |       |

Answer

$$\begin{aligned} \frac{d}{dx} \tan^3(5x+4) &= \frac{d}{dx} [\tan(5x+4)]^3 \\ &= 3 [\tan(5x+4)]^2 \cdot \sec^2(5x+4) \cdot 5 \\ &= 15 \tan^2(5x+4) \sec^2(5x+4) \end{aligned}$$

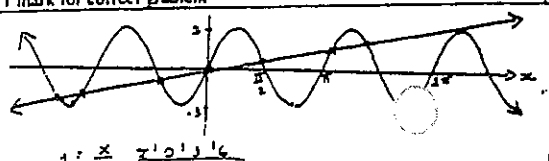
(b)

(i) Outcomes assessed: P4 H5

Marking Criteria

| Criteria                            | Marks |
|-------------------------------------|-------|
| • 1 mark for sine graph             | 4     |
| • 1 mark for correct range & period |       |
| • 1 mark for straight line graph    |       |
| • 1 mark for correct gradient       |       |

Answer



| Criteria                                         | Marks |
|--------------------------------------------------|-------|
| • 1 mark for linking given equation to the graph | 2     |
| • 1 mark for correct answer according to graph   |       |

Answer (i)  $6 \sin 2x - x = 0$  Since  $y = 3 \sin 2x$  and  $y = \frac{x}{2}$   
 $\text{ie } \frac{6 \sin 2x}{2} = \frac{x}{2}$  intersect 7 times  
 $\text{ie } 3 \sin 2x = \frac{x}{2}$   $6 \sin 2x - x = 0$  has 7 solutions

QUESTION 8

(a) Outcomes assessed: P5 H6

Marking Criteria

| Criteria                                        | Marks |
|-------------------------------------------------|-------|
| • 1 mark for 1st and 2nd derivatives            | 3     |
| • 1 mark for correct link to increasing curve   |       |
| • 1 mark for correct link to concave down curve |       |

(b) Outcomes assessed: P7 H5

| Criteria                    | Marks |
|-----------------------------|-------|
| • 1 mark for 1st derivative | 3     |
| • 1 mark for 2nd derivative |       |
| • 1 mark for proof          |       |

Answers

(a)  $\frac{dy}{dx} = \frac{1}{x}$

$\frac{d^2y}{dx^2} = -\frac{1}{x^2}$

When  $x > 0$ ,  $\frac{1}{x} > 0 \therefore$  INCREASING

$-\frac{1}{x^2} < 0 \therefore$  CONCAVE DOWN

Hence  $y = \log_e x$  is concave down + increasing for all  $x > 0$

(c) Outcomes assessed: P8 H8

| Criteria                                    | Marks |
|---------------------------------------------|-------|
| • 1 mark for identifying the integral       | 3     |
| • 1 mark for integration                    |       |
| • 1 mark for substitution to correct answer |       |

(d) Outcomes assessed: P8 H5

| Criteria                    | Marks |
|-----------------------------|-------|
| • 1 mark for integration    | 3     |
| • 1 mark for substitution   |       |
| • 1 mark for correct answer |       |

Answers

(c)  $\int_1^b \frac{1}{x} dx = 1$  ✓

$[\log_e x]_1^b = 1$

$\log_e b - \log_e 1 = 1$

$\log_e b - 0 = 1$

$\therefore b = e$

OR  $\int_b^1 \frac{1}{x} dx = 1$

$[\log_e x]_b^1 = 1$

$\log_e 1 - \log_e b = 1$

$0 - \log_e b = 1$

$\therefore b = e^{-1}$

(d)  $\frac{dy}{dx} = 2 \sin 3x$

$y = -\frac{2}{3} \cos 3x + C$  ✓

$\frac{\pi}{2} = -\frac{2}{3} \cos(3 \cdot \frac{\pi}{2}) + C$

$\frac{\pi}{2} = -\frac{2}{3} \cos(\frac{3\pi}{2}) + C$

$\frac{\pi}{2} = -\frac{2}{3} \cdot 1 + C$

$C = \frac{\pi}{2} + \frac{2}{3}$

$\therefore y = -\frac{2}{3} \cos 3x + \frac{\pi}{2} + \frac{2}{3}$  ✓

(1)

7(a)  $F(x) = \frac{x+1}{x-1}$ ,  $G(x) = x^2$ ,  $H(x) = \sqrt{x}$  (i) Domain  $F(x)$   $\forall x \in \mathbb{R}$  except  $x=1$

(ii)  $H(G(-3)) = H(9) = 3$ .  $G(H(-3)) = G(\sqrt{-3})$  But  $\sqrt{-3}$  is not a Real Number  $\therefore$  Value  $\sqrt{-3}$  not useable.

(iii)  $F(\frac{1}{x}) = \frac{\frac{1}{x}+1}{\frac{1}{x}-1}$   $\therefore G(\frac{1}{x})$  not possible. (1)

$$= \frac{1+x}{1-x} \text{ or } -\left(\frac{1+x}{x-1}\right) \therefore F(x) = -\left(\frac{1+x}{x-1}\right)$$

$$\therefore F\left(\frac{1}{x}\right) = -F(x) \quad (2)$$

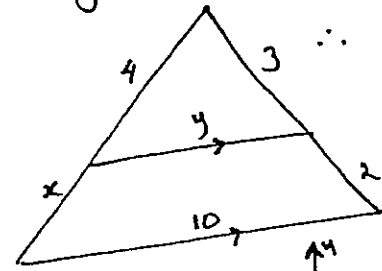
(iv) For inverse of  $F(x)$  rewrite as  $y = \frac{x+1}{x-1}$  & for inverse  $(x, y)$  is  $x = \frac{y+1}{y-1}$

$$\therefore x(y-1) = y+1 \text{ & } xy - y = x+1$$

So  $y(x-1) = x+1$  or  $y = \frac{x+1}{x-1}$  But this  $y$  is  $f^{-1}(x)$  (3)

$$\therefore F^{-1}(x) = \frac{x+1}{x-1}$$

(b)

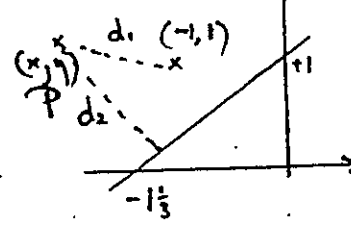


From the diagram & considering similar  $\Delta$ s

$$\frac{4}{x} = \frac{3}{2} \Rightarrow x = \frac{8}{3} \text{ or } 2\frac{2}{3} \quad (2)$$

$$\text{& } \frac{y}{10} = \frac{3}{5} \Rightarrow y = 6. \quad (2)$$

Q10 (a)  
4 marks



Let  $d_1$  be distance from  $P$  to  $(-1, 1)$  is  $d_1 = \sqrt{(x+1)^2 + (y-1)^2}$

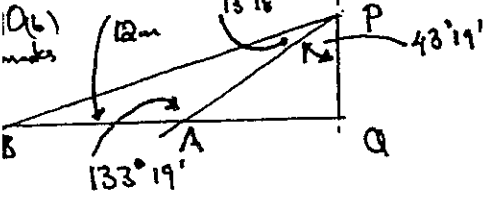
& let  $d_2$  be distance  $P$  to line is  $d_2 = \frac{|3x-4y+4|}{\sqrt{9+16}} \quad (1)$

$$\text{But } d_1 = d_2 \Rightarrow d_1^2 = d_2^2 \Rightarrow (x+1)^2 + (y-1)^2 = \left(\frac{3x-4y+4}{5}\right)^2 \quad (1)$$

$$25(x^2+2x+1) + 25(y^2-2y+1) = (3x-4y+4)^2$$

$$25x^2+50x+25 + 25y^2-50y+25 = 9x^2+24x+16y^2-32y-24xy+16$$

So the required equation of the locus is  $16x^2+26x+24xy+9y^2-18y+34 \quad (1)$

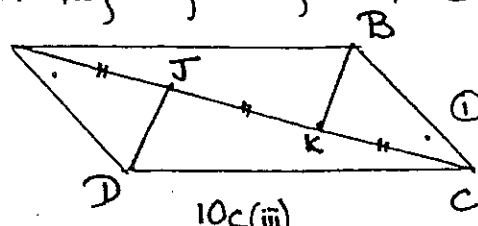


$$\frac{BP}{\sin 133^\circ 19'} = \frac{12}{\sin 13^\circ 16'} \quad (2)$$

$$\therefore BP = 37.95213536$$

$$\text{ie } BP = 37.95 \text{ m} \quad (1)$$

10(c) A  
Similar



10(c)(ii)

Consider  $\Delta ADJ$  &  $\Delta CBK$

$$AJ = CK \text{ given}$$

$$\angle DAJ = \angle BCK \text{ alt } \angle \text{ AD // BC}$$

$$(3) \quad AD = BC \text{ opp sides of } \Delta$$

$$\therefore \Delta ADJ \equiv \Delta CBK \text{ SAS}$$

$$\therefore \angle AJD = \angle CKB \text{ Corresponding } \angle \text{ s Congruent } \Delta$$

$$\text{Since } \angle AJD = \angle CKB$$

$$\text{then } \angle DJK = \angle BKT$$

$$\text{Adj } \angle \text{ s on st line}$$

$$\therefore DJ // BK$$

$$\text{Alternate } \angle \text{ s equal.}$$

(1)